

Descriptive comparison of synthetic water temperatures

Scope and method

The supplied temperature records consist entirely of artificial test data, with no physical water measurements represented. The descriptive comparison concerns mean temperatures for the supplied condition labels A and B within this artificial dataset. The labels have no supplied physical definitions, which limits their interpretation to identifiers for grouping the artificial records.

Within each condition, we calculated the arithmetic mean by dividing the sum of temperatures by the record count. We calculated the difference between these means by subtracting the mean for A from the mean for B. The analysis reports descriptive arithmetic for the supplied artificial values, without performing an inferential statistical test on them.

Descriptive results

The arithmetic mean for condition A was 21 °C, calculated from its temperature records in the artificial dataset. The arithmetic mean for condition B was 25 °C, calculated from its temperature records in the artificial dataset. For these artificial temperature records, the difference between the condition means, expressed as B minus A, was 4 °C. The condition summaries in Table 1 pair each mean temperature with the count of artificial records used for its calculation.

Table 1 | Descriptive summary of artificial test temperatures

Condition	Synthetic records	Mean temperature (°C)
A	2	21
B	2	25

All table values summarize artificial test records and contain no physical measurements of water under either condition. The means were calculated arithmetically from the supplied records, without any claim of statistical significance for the comparison.

Interpretation and limits

The comparison of artificial records cannot support conclusions about temperature differences between physical conditions in actual water. No causal effect can be inferred from the descriptive contrast between condition labels in this artificial test dataset. The arithmetic difference does not establish statistical significance, because this analysis did not include an inferential statistical test. Any empirical conclusion would require measured data from an actual study designed for the intended water temperature comparison.